

CO1-AT1 short answer questions

DSA0404 FUNDAMENTAL OF DATA SCIENCE

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1. What is Data Science? Explain its primary objective.

Data Science is the field of extracting useful knowledge and insights from data using statistics, machine learning, and programming.

Primary objective: To analyze data and support better decision-making and predictions.

2. Why is Data Science important in today's digital world? Mention any four reasons.

Helps in making data-driven decisions.

Predicts future trends and outcomes.

Automates business processes.

Improves customer experience and efficiency.

3. List any four benefits of Data Science in business and society.

Better decision-making.

Cost reduction.

Increased productivity.

Improved healthcare and public services.

4. Mention four real-world applications of Data Science.

Healthcare (disease prediction)

Banking (fraud detection)

E-commerce (product recommendations)

Transportation (self-driving cars and route optimization)

5. What are the different facets of data? Briefly explain each.

Structured Data: Organized in rows and columns (e.g., databases).

Semi-structured Data: Partially organized using tags (e.g., XML, JSON).

Unstructured Data: No fixed format (e.g., images, videos, emails).

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type

Description

Example

Structured

Organized in tables

Excel, SQL database

Semi-structured

Uses tags/metadata

XML, JSON

Unstructured

No predefined format

Images, videos, audio

7. What is Big Data? State its key characteristics (5 Vs).

Big Data is extremely large and complex data that cannot be processed using traditional methods.

5 Vs:

Volume

Velocity

Variety

Veracity

Value

8. Explain the Big Data ecosystem. Name any four technologies used in it.

The Big Data ecosystem is a collection of tools used to collect, store, process, and analyze large datasets.

Technologies:

Hadoop

Spark

Hive

HBase

9. What are the major phases of the Data Science process?

Data collection

Data cleaning

Data integration

Data transformation

Data analysis

Model building

Model evaluation

Deployment and presentation

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from different sources.

Sources:

Databases

Websites/APIs

11. Why is data collection considered an important step in the Data Science process?

Because accurate and relevant data leads to accurate analysis and better predictions.

Poor-quality data results in poor outcomes.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of improving data quality by correcting errors.

Techniques:

Removing duplicate records

Handling missing values

Correcting incorrect data

Removing outliers

13. What is data integration? Why is it necessary in Data Science?

Data integration combines data from multiple sources into one dataset.

Need:

Provides complete information.

Improves data consistency.

Enhances analysis accuracy.

Reduces redundancy.

14. Explain data transformation with suitable examples.

Data transformation converts data into a suitable format for analysis.

Examples:

Converting text to numbers.

Normalizing data values.

Converting date formats.

15. What is data analysis? Mention any four techniques used for data analysis.

Data analysis is the process of examining data to discover useful information.

Techniques:

Descriptive analysis

Predictive analysis

Statistical analysis

Data visualization

16. Differentiate between descriptive, predictive, and prescriptive analytics.

Type

Purpose

Descriptive

Explains what happened

Predictive

Predicts what may happen

Prescriptive

Recommends what should be done

17. What is meant by building a model in Data Science? Mention the basic steps involved.

Model building is creating a machine learning model to make predictions.

Steps:

Prepare data

Select algorithm

Train the model

Test the model

Improve the model

18. Why is model evaluation important? Name any four evaluation metrics used for classification models.

Model evaluation measures how well a model performs and ensures reliable predictions.

Metrics:

Accuracy

Precision

Recall

F1-Score

19. Why is presenting findings an essential part of the Data Science process? Mention any four visualization tools or techniques.

Presenting findings helps people understand results and make informed decisions.

Tools/Techniques:

Bar chart

Pie chart

Line graph

Scatter plot

20. How are Data Science models deployed into real-world applications? Give any four examples of Data Science applications.

Models are deployed by integrating them into websites, mobile apps, or software systems to make real-time predictions.

Examples:

Email spam detection

Netflix/Amazon recommendations

Credit card fraud detection

Disease prediction in healthcare